

PRECAMBRIAN				
AGE (Ma)	EON	ERA	PERIOD	BDY. AGES (Ma)
541	PROTEROZOIC	Multi-cell	④ EDIACARAN	541
635			CRYOGENIAN	635
720		NEOPROTEROZOIC	TONIAN	720
1000				1000
1250		MESOPROTEROZOIC	STENIAN	1250
1400			ECTASIAN	1400
1600			CALYMMIAN	1600
1750		PALEOPROTEROZOIC	STATHERIAN	1750
1800			OROSIRIAN	1800
2050				2050
2250			RHYACIAN	2250
2300	ARCHEAN	③ SIDERIAN Photosyn.		2300
2500				2500
2750		NEOARCHEAN		2750
2800		MESO-ARCHEAN		2800
3200				3200
3600		PALEO-ARCHEAN		3600
3750	EOARCHEAN		② Single cell	3750
4000			① life	4000
	HADEAN			

1) Life starts ~ 3.8 bya

- Life Began in hydrothermal vents under ocean (Black smokers)
- Released dangerous rich minerals in Ocean (100°, toxic metals, O₂)
- Vents supported **chemosynthesis** for bacteria by providing sulfur/other compounds, etc
- Life relied on chemical energy w/o photosynthesis

2) Oldest known fossils from Australia

- Tells us early consisted of simple, single-celled organisms forming microbial communities
- Their existence implies life originated earlier than this since organisms need time to evolve before leaving fossil fuels

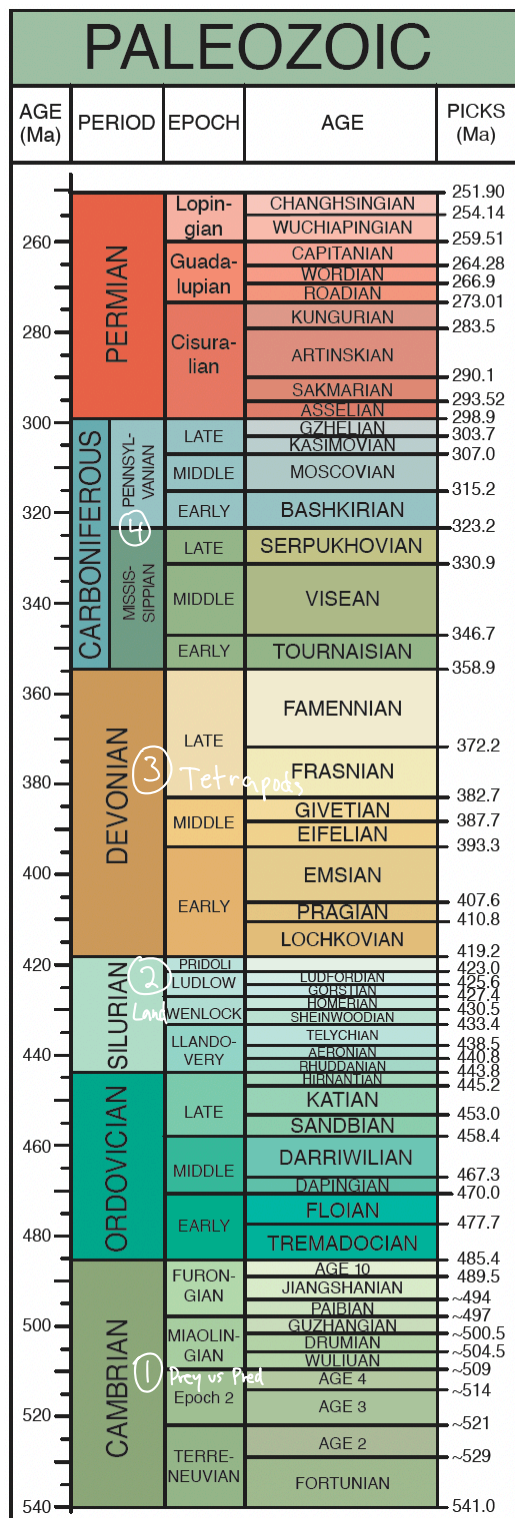
3) Origin of Oxygenic Photosynthesis (~2.5 bya)

- P.S. first evol from **cyanobacteria** (similar to anabaena)
 - Sun + water + CO₂ → Glucose + Oxygen (O₂)
- P.S. significantly increased O₂ levels leading to **Great Oxidation event** over time
- P.S. enabled **aerobic respiration** that was way more eff @ producing cell energy;
 - Its eff lead to huge evol advantages/complex organisms

4) End of Precambrian Era

- We find fossils of large multicellular organisms
 - Soft-bodied, flat, mostly marine, filter feeders
- Most of the organism have no clear modern relatives

Start: single cell → Photosyn: Multicell



1) Cambrian Explosion (540-490 mya)

- Period of rapid animal diversification
- Most major phyla appear here
- Organisms develop complex bodies
- Organisms get nutrients from prey now instead of water/sediment
- Created Evol pressures
- We see claws, jaws, spines, armor + rapid movement

2) Silurian Period: Moving to Land

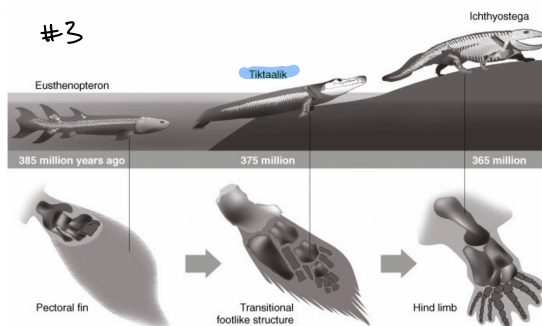
- Multi-celled orgs moved to land but **faced challenges**
 - Dehydration
 - Nutrients w/o dissolved water
 - Respiration / gas exchange
 - Reproduction w/o water env
 - Gravity / Locomotion (body weight vs gravity)
 - Adapting sensory system to air
- Despite these land offer **evol advantages!**
 - Early terrestrial:
 - little competition, predation, disease
 - Little comp = new ecological opportunities → more evol
- Terrestrial Ecosystems were forming (450-420 mya)
 - **Cooksonia**: First land plant;
 - lacked roots/leaves
 - Relied on mycorrhizal fungi for nutrients
 - These two jump started land colonization

3) Devonian Period: Tetrapods Intro (Amphibians, Reptiles, Mammals)

- Aquatic → Terrestrial vertebrates
- **Tiktaalik Fossil**
 - Shows link from fish → tetrapods
 - Has limbs for **supporting body weight** (vs fins for swimming)
 - Suggest it can move in shallow water/shores
- Scientist studied rock layers + evol timelines to predict where

4) Carboniferous Period: O₂ + Giant Bugs

- Highest O₂ levels in history
- O₂ levels create diverse forests + allow organism to grow much larger through diffusion
- A lot of fossil created from these organisms



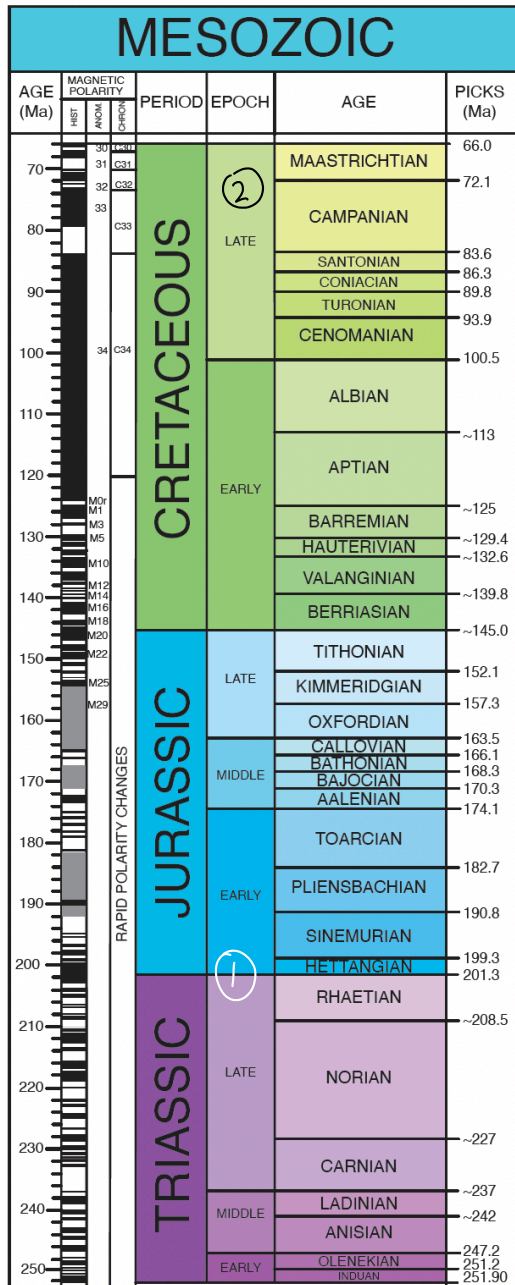
Paleozoic:

"University of Cambrian Predators"

"Silu moves to land" (He faces challenges, but evols)

"Devonian Mitchell likes Tetrapods"

"Carboniferous Trees + Bugs"



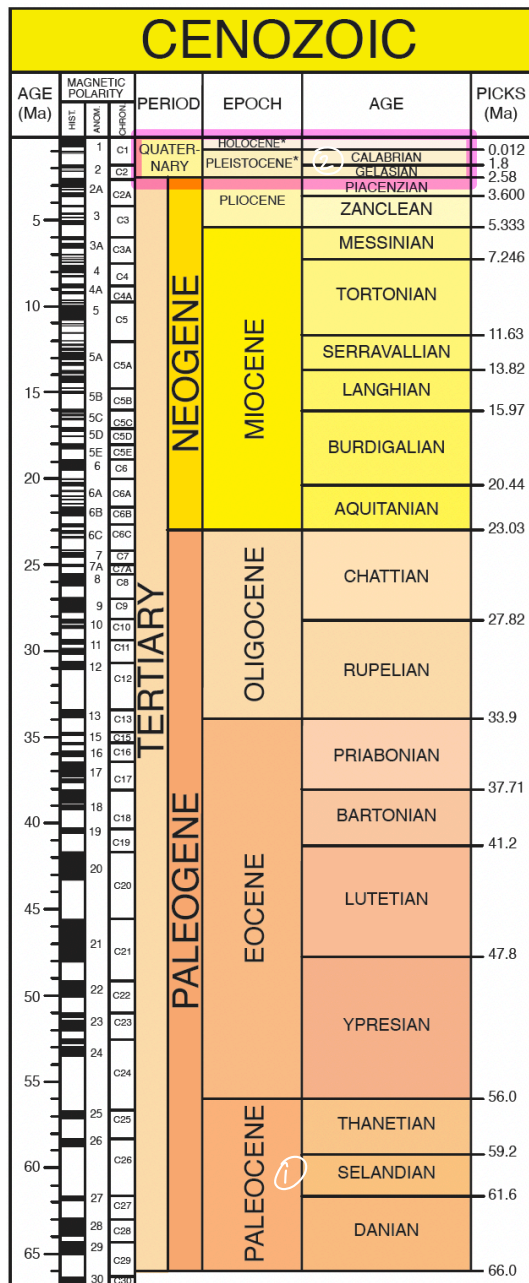
1) Mesozoic : Dinosaur Age

- Triassic Period : First dino
- Jurassic Period : Dinosaurs diversified
- Cretaceous Period : Dino's dominate ecosystem.
- Birds evolved from dino lineage
- Only surviving dino's today

2) Dino Extinction Event (~65 mya)

- Evidence suggest cause was large asteroid impact near Gulf of Mexico (left a 110 mile wide crater)
- Left dust/debris in atmos blocking sun / disrupting ecosystems
- Evidence includes :
 - Global Iridium + microtektites in rock layers

Dinosaur = Meso - Saur Power Rangers



1) Early Paleogene : Rise of Mammals

- Mammals rapid overtook dino eco niches after extinction
- Mammal existed before, but w/ this many new mammal lineages began appearing and branching out
- We see rapid diversification early Cenozoic

2) Quaternary Period : Human Origins

- Homo sapiens first evolved in Africa
- They lived in Africa for 10s of thousands of years
- 60-70K years ago humans started migrating
- Europe → Asia → Australia → Americas
- New env → adapted to : climate, landscape = diverse human cultures + lifestyles
- Migration Patterns Evidence :
 - Fossils : where humans lived
 - Stone tools : Show tech dev ; can date to locations / time
 - Foot prints : Evidence from sediments ; shows human movement
 - Cultural Artifacts : Like beads / ornaments indicate behavior + help track how human culture spread over time

Cenozoic :

"Dinos gone? Humans In"

"Migrating Humans..."

"Present ..."